

AN APPROACH TO OPTIMAL DISTRIBUTION SYSTEM PLANNING THROUGH CONDUCTOR GRADATION

M. PONNAVAIKKO K.S. PRAKASA RAO

Department of Electrical Engineering and
Centre of Energy Studies
Indian Institute of Technology
New Delhi 110 016
INDIA

Abstract - An optimal conductor gradation procedure for radial distribution feeders is proposed in this paper. Models to represent feeder cost, energy loss cost and voltage regulation as a function of conductor cross-section and an objective function for optimizing the conductor cross-section have been formulated. The problem has been posed as a multistage decision dynamic programming problem, and the optimal conductor cross-sections for the feeder segments are obtained by solving the same. The proposed method takes into account the non-uniform distribution of loads along the length of the feeders and also the maximum permissible voltage drop along a feeder. Further, the method takes into account the load growth in future years of the plan period. The results of the design of a typical radial feeder using the proposed method are presented and discussed.

INTRODUCTION

The distribution system constitutes a significant part of a total power system. As can be seen from the investment pattern of several electrical utilities, the capital deployed in the distribution system forms about 40% of the total amount spent in the entire power system. Also, since the distribution voltage level is low and the low tension distribution system is extensive, this subsystem is prone to have more losses compared to that in the other parts of the system, sometimes constituting as high as 80% to 85% of the overall losses in the entire power system. Moreover, the energy cost will be maximum at the distribution level and a loss of the same at the distribution level will lead to more severe financial implications. It is, therefore, essential to minimize the overall cost of the distribution system including the capital investment and the present worth of the cost of energy losses in the life span of the system through an optimal design.

Optimal distribution system design has gained considerable importance in the recent past [1,2,3], but in all these attempts, the radial feeders have been assumed to be of uniform cross-section throughout their lengths. In radial feeders, the line segments closer to the source carry the maximum load while the line segments far away from the source are lightly loaded. This inherent characteristic of the radial feeders enables the choice of multiple conductor cross-sections for a single feeder. This would lead to a reduction in the system cost compared to that with a single conductor cross-section. At the same time, use of a large

number of conductors of different cross-section, will result in increased cost of the inventory. A judicious choice can, however, be made in the selection of number and size of conductor cross-sections for consideration in the optimal design.

The problem of conductor gradation has been studied through an enumeration technique in the past by A.W. Funk Houser [4] for distribution feeders, assuming uniform load distribution only. Also, he combined the phase value of the accumulated cost of the energy losses occurring in the later periods with the present investment on the conductors which may lead to non-optimal solutions, since the present worth of the energy loss cost at a later date is much less than its phase value at the time of its loss.

Mathematical models for voltage drop, feeder cost and the present worth of energy loss cost taking load growth into consideration have been developed in this investigation based on the models developed in [3] and an attempt has been made in this work to suggest a solution procedure for obtaining optimal conductor gradation for radial feeders using Dynamic Programming Approach [5,6]. The technique, presented, is independent of circuit voltage, load distribution and other system and load characteristics and thus, is applicable to all types of radial feeders in practice.

PROBLEM FORMULATION

Formulation of Mathematical Models

Mathematical models to represent feeder voltage drop, energy loss cost and feeder cost, as a function of conductor cross-section, are developed first to facilitate the choice of an objective function, needed.

Feeder voltage drop: It is known from [3] that the approximate voltage drop, v , in volts in a radial feeder with n segments, feeding loads with lagging power factor can be obtained as

$$v = \sum_{i=1}^n (\sqrt{3}) (I_i R_i \cos \theta + I_i X_i \sin \theta) \quad (1)$$

From Ohm's Law, the resistance in ohms per phase of the i th line-segment of the feeder is given by

$$R_i = \frac{\rho l_i}{a_i} \quad (2)$$

The reactance in ohms per phase of the i th line-segment of the feeder is obtained from (3)

$$X_i = x l_i \quad (3)$$

where x is the per phase reactance in ohms per km of the feeder with a given conductor size. The value for x of the distribution lines with different ACSR conductors ranging from Squirrel to Dog remains almost constant. For example, with ACSR conductors ranging from Squirrel to Dog, the per phase reactance, x in ohms per km lies between 0.32 and 0.29 for 0.4 kV distribution lines and between 0.39 and 0.36 for 11 kV

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distribution lines. The effect of this narrow range of variation in x of the lines with different conductor sizes on the voltage drop is very small and hence can be neglected in the analysis. Thus, keeping x constant, the equation for voltage drop in the feeder can be obtained from (1) and (2) as

$$v = \sum_{i=1}^n \left(\frac{k_{1i}}{a_i} + k_{2i} \right) \quad (4)$$

where

$$k_{1i} = \sqrt{3} I_i \ell_i \rho \cos \theta \quad (5)$$

$$k_{2i} = \sqrt{3} I_i \ell_i x \sin \theta \quad (6)$$

Energy loss cost: The energy loss in kWh in the feeder with n line-segments for the loading conditions in the base year (0 th year) is given by

$$EL = \sum_{i=1}^n 26.28 I_i^2 R_i (LLF) \quad (7)$$

where the loss load factor, LLF is a function of load factor, LF and is defined [3] as

$$LLF = A (LF)^2 + B (LF) \text{ and } A+B = 1 \quad (8)$$

In the analysis, the values for A and B are taken as 0.8 and 0.2, respectively. The cost of energy loss is a recurring expenditure annually to the utilities throughout the life period of the feeders. The sum of the present worth of the cost of energy losses in the life period of N years at a discount rate of r , can be obtained as

$$C_{Po} = \sum_{i=1}^n 26.28 I_i^2 R_i (LLF) c \sum_{k=1}^N \frac{1}{(1+r)^k} \quad (9)$$

where c is the cost of energy per kWh. As could be seen from (9) the energy loss cost is a function of load current, loss load factor, and cost of energy. Variation of any of these factors will change the cost of energy loss and hence will affect the optimal conductor gradation. Therefore, the model should consider the effect of load growth, growth in load factor and change in cost of energy.

Effect of load growth: Load growth in an area with time is a natural phenomenon. The growth in feeder load may be due to addition of new loads to the feeder or due to the incremental additions to the existing loads. A feeder designed and constructed in a system on long-term basis can accept additional loads to the extent it can accommodate satisfying the voltage constraints. Once the load exceeds the feeder capacity, limited by either voltage regulation or thermal constraints, new facilities such as substations or additional feeders need to be created. Till such time the substation feed area and the configuration of the main feeders may be assumed to remain constant. It is further assumed that the feeder load grows at a predetermined annual rate, g , in proportion to the loads connected at the tapping points, implying that the feeder load profile, on a per-unit basis, remains unchanged with load growth. This assumption is made for want of sufficient information on the exact pattern of load growth along the feeder main. However, the effect of this assumption on optimal conductor gradation is negligible, as experienced by the authors. The effect of load growth on the annual cost of energy losses can be introduced by multiplying (9)

by a factor $(1+g)^{2k}$, $k=1, \dots, M$, where M is the plan period upto which the feeder can take load growth.

Effect of growth in load factor: The system experienc-

es growth in load factor due to many reasons such as increase in load diversity with load growth, increase in energy consumption per kW connected load with time, measures taken by the utilities to flatten load curves for improving system efficiency and to curb the growth in peak demand etc. According to Scheer[7], the system load factor grows cutting the difference between an ultimate load factor and the present load factor into half over a period of 16 years, and the load factor at any year, k , is given by

$$LF_k = LF_u - y_k (LF_u - LF_p), \text{ where } y_k = (0.5)^{k/16} \quad (10)$$

The corresponding annual loss load factor, LLF_k , $k=1, \dots, N$, can be obtained from (8) and substituted in (9) for considering the effect of increase in load factor in the model. However, since the load growth on the feeder is limited to M years, the increase in LF due to other factors beyond M years are ignored.

Effect of cost of energy: The cost of equipments, construction and maintenance increase with time. This results in a continuous increase in the cost of energy with time. As the increase in cost of energy will have a definite influence on the optimal conductor gradation, the same may be reflected into the model by introducing the variable cost of energy, c_k , $k=1, \dots, N$ in (9).

Thus, considering the effect of change in load, load factor and cost of energy the present worth of the cost of energy losses over the life of the system can be obtained from (9) and (2) as in (11).

$$C_{PN} = \sum_{i=1}^n \frac{k_{3i}}{a_i} \quad (11)$$

$$\text{where } k_{3i} = 26.28 \rho \ell_i I_i^2 \sum_{k=1}^M \frac{(1+g)^{2k} (LLF)_k c_k}{(1+r)^k} + (1+g)^{2M} (LLF)_M \sum_{k=M+1}^N \frac{c_k}{(1+r)^k} \quad (12)$$

Feeder cost: The feeder cost is of the form

$$C_b = e + f a \quad (13)$$

where a is the area of conductor cross-section, e and f are constants to be determined from cost versus conductor cross-section characteristic of the line. Typical values in an Indian Power System for e and f in the case of 11 KV feeders are Rs.5000 and Rs.260 per mm²/km respectively. The cost of the feeder with n line-segments can be represented in the form

$$C_f = \sum_{i=1}^n (k_{4i} a_i + k_{5i}) \quad (14)$$

$$\text{where } k_{4i} = f \ell_i \text{ and } k_{5i} = e \ell_i \quad (15)$$

Objective Function

The objective function, to be minimized, is defined assuming a fixed feeder configuration. Since minimization of feeder cost alone may lead to more energy losses and on the other hand minimization of energy loss cost independently may result in increased feeder cost, the objective function should be a function of both the feeder cost and the energy loss cost.

The term $\sum_{i=1}^n k_{5i}$ in (14) may be dropped while

defining the objective function, since this term is independent of conductor cross-section.

The main consideration in the feeder design is to keep the voltage drop within a prespecified limit. This limit normally depends on the system requirements to maintain the declared voltage to the customers and on the economic criteria. Thus, the choice of this maximum li-

mit of voltage drop should be based on the optimal voltage regulation obtained through overall system optimization studies [3] for the area in which the feeders, under investigation, are serving.

Another constraint is that the conductor cross-section a_i , cannot take any value, due to the non-availability of conductors of any size in the inventory and is constrained to take only discrete values, corresponding to the available conductor cross-sections in the inventory.

The problem can, thus, be stated as

$$\text{Minimize } Z = \sum_{i=1}^n \left(\frac{k_{3i}}{a_i} + k_{4i} a_i \right) \quad (16)$$

$$\text{subject to } \sum_{i=1}^n \left(\frac{k_{1i}}{a_i} \right) \leq b \quad (17)$$

and

$$a_i > 0 \text{ (taking discrete values)} \quad (18)$$

$$\text{where } b = D - \sum_{i=1}^n k_{2i}, \quad (19)$$

D being the maximum allowable voltage drop for the base year loading condition.

It may be seen from (17) or (19) that they are independent of load growth. This is evident from the assumptions made on load growth along the feeder main and design considerations as discussed under the section on effect of load growth. The voltage drop b or D in (17) or (19) corresponds to the present load. As the feeder is designed to meet load growth, the allowable voltage drop at an year k will be $(1+g)^k D$ and the coefficients k_{1i} in (17) and k_{2i} in (19) will respectively be $(1+g)^k k_{1i}$ and $(1+g)^k k_{2i}$. Thus, the term $(1+g)^k$ on both RHS and LHS of (17) or (19) gets cancelled, thus, making the constraint equation independent of load growth. The limitation of conductor gradation and other considerations are discussed in Appendix B.

SOLUTION APPROACH

Dynamic Programming

Large number of radial feeders are there in service in a distribution system and the load distribution in them is different from each other and is mostly non-uniform in practice. It is, therefore, difficult to formulate generalised guide lines for the optimal grading of conductors for the radial distribution feeders. Also, since every feeder in the system needs to be analysed individually, enumeration technique is not employed due to the requirement of massive computational efforts. It is, therefore, necessary to look for a simpler optimal grading method requiring reasonably less computational effort.

A closer examination will reveal that the present problem can be posed as a dynamic programming problem. As stated in (16) and (17), it is required to minimize a separable cost function with respect to the variable conductor cross-sections subject to the voltage drop constraint. The terminals of the line segments can be identified as the different stages and at each stage, the cumulative voltage drop that can be allowed starting from the far end may be considered as the state variable. This variable can assume any value in the allowable range subject to a certain minimum and a maximum value. In each line segment, the conductor cross-section can be chosen from out of a standard set of conductor cross-sections available in the inventory and the conductor cross-sections can be viewed as the decision variables and these are discrete variables. The voltage drop in a line segment depends on the choice of a

conductor cross-section and, the value of the objective function, in turn, depends on the choice of conductor cross-section. The allowable voltage drop can thus, be treated as a resource to be allocated to the different line-segments through an optimal choice of conductor cross-sections that will minimize the objective function. The transition from any feasible state at any stage to another feasible state in the next stage can be effected through the choice of a decision variable (conductor cross-section).

The problem can be written as

$$\min_{a_1 \dots a_n} Z = \sum_{i=1}^n f_i(a_i) \quad (20)$$

subject to

$$\sum_{i=1}^n (k_{1i}/a_i) \leq b \quad (21)$$

and

$$a_i > 0, \text{ (taking discrete values)}$$

defining the parameters under the framework of dynamic programming as follows :

The decision variables are the conductor cross-sections a_i , each of which can assume a set of discrete values corresponding to the conductors available in the inventory.

The stage return functions are

$$f_i(a_i) = \frac{k_{3i}}{a_i} + k_{4i} a_i \quad (22)$$

The contribution of voltage drop from the individual line segments, d_i is as in (23).

$$d_i = \frac{k_{1i}}{a_i} \quad (23)$$

The state variables are

$$S_n = \sum_{i=1}^n d_i \leq b$$

$$S_{n-1} = \sum_{i=1}^{n-1} d_i = S_n - d_n = t_n(S_n, a_n)$$

⋮

$$S_2 = \sum_{i=1}^2 d_i = S_3 - d_3 = t_3(S_3, a_3)$$

$$S_1 = d_1 = S_2 - d_2 = t_2(S_2, a_2) \quad (24)$$

The stage transition function is, thus, given as

$$S_{i-1} = t_i(S_i, a_i) \quad (25)$$

Each of the state variables S_i , $i = 1, \dots, n$, can take all the values within a feasible range, defined by a minimum and a maximum bounds. The minimum bound is given by

$$S_{i \min} = \sum_{r=1}^i d_{\min, r} \quad (26)$$

where $d_{\min, r}$ is the minimum voltage drop contribution

in the r th line segment with the largest conductor cross-section. This assumption ensures feeder conti-

nunity and keeps the search area minimum.

The maximum bound, $S_{i_{\max}}$ is the maximum permissible voltage drop, b .

State functions, $F_i(S_i)$ are defined to represent the minimum value of the objective function up to stage i , as

$$F_i(S_i) = \min_{a_i - a_1} \sum_{j=1}^i f_j(a_j) \quad (27)$$

The state functions $F_i(S_i)$, $i = 2, \dots, n$ can be obtained from the recursive relations as in (28)

$$F_i(S_i) = \min_{a_i} \left\{ f_i(a_i) + F_{i-1}[t_i(S_i, a_i)] \right\} \quad (28)$$

and for the stage 1,

$$F_1(S_1) = \min_{a_1} [f_1(a_1)] \quad (29)$$

and, thus, the optimal value of the objective function is given by

$$Z^* = \min_{a_n} \left\{ f_n(a_n) + F_{n-1}[t_n(S_n, a_n)] \right\} \quad (30)$$

The optimal values of a_i which minimize $F_i(S_i)$ for every feasible value of S_i at stage i , $i = 1, \dots, n$ are defined as policy function, $\hat{a}_i(S_i)$.

Always, the minimum value of the objective function in the final stage occurs when S_n takes the maximum value, b , and thus

$$Z^* = F_n(b) \quad (31)$$

and

$$a_n^* = \hat{a}_n(b) \quad (32)$$

After determining a_n^* for the n th line segment at the n th stage, $a_1^*, \dots, a_{n-1}^*$ can be obtained from the policy functions $\hat{a}_i(S_i)$ using (33)

$$a_{n-1}^* = \hat{a}_{n-1}(b - \sum_{r=0}^{i-1} \frac{k_{1,n-r}}{a_{n-r}}), i=1, \dots, n-1 \quad (33)$$

The method is illustrated through a simple example in the Appendix A.

ALGORITHM DESCRIPTION

The solution approach discussed in the preceding section to solve the conductor gradation problem by defining the same as an n -stage decision problem under the framework of dynamic programming, requires the computation of the coefficients in the objective and constraint functions, to start with, which involves the estimation of peak load current flows in the line segments. The voltage magnitudes at the terminals can be obtained by distributing the maximum allowable voltage drop in proportion to the kVA-km flow in the line segments, from which the current flows in the line segments can be computed.

The steps of solving the problem using the proposed algorithm are as follows:

Steps

1. Compute the coefficients in (16) and (17) and estimate b from (19).
2. Compute the stage returns and the voltage drop contributions of the individual line-segments from (22) and (23), respectively for every value of a_i available in the inventory.
3. Start with stage 1 and compute $F_1(S_1)$ from (29) for all the feasible values of S_1 between $S_{1_{\min}}$ and $S_{1_{\max}}$.
4. Generate the policy function $a_1(S_1)$ corresponding to $F_1(S_1)$ computed in step 3.
5. For stages from 2 to $n-1$, compute $F_i(S_i)$, $i=2, \dots, n-1$, using the recursive relations in (28) and generate the corresponding $\hat{a}_i(S_i)$, $i=2, \dots, n-1$, for every feasible value of S_i between $S_{i_{\min}}$ and b .
6. At stage n , compute $F_n(b)$ and obtain $\hat{a}_n(b)$ which give Z^* and a_n^* respectively as in (31) and (32).
7. Determine $a_1^*, \dots, a_{n-1}^*$ from (33) tracing $\hat{a}_i(S_i)$ back from the $(n-1)$ th stage to the first stage to obtain the optimal conductor gradation along the feeder.

DISCUSSION OF RESULTS AND CONCLUSION

The proposed method has been tested on a number of 11 kV radial distribution feeders in an Indian distribution system designed with single conductor configuration. The results of one such feeder is presented here for discussions. The loading pattern of the feeder is shown in Fig.1. The other data assumed in the studies are given in Table 1.

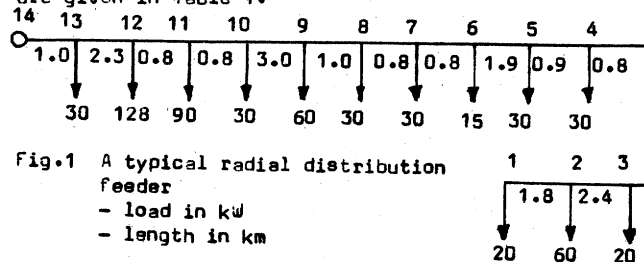


Fig.1 A typical radial distribution feeder
- load in kW
- length in km

Table 1
Data

1. Resistivity of Aluminium	: 28.1 ohm-mm ² /km
2. Constant reactance for the distribution lines	: 0.38 ohm/km
3. Present load factor	: 0.2
4. Ultimate load factor	: 0.45
5. Annual load growth rate	: 0.1
6. Load growth period	: 10 years
7. Discount rate	: 0.1
8. Life of the feeder	: 25 years
9. Power factor	: 0.3
10. Cost of the line (variable part)	: Rs.260/mm ² /km
11. Energy cost (assumed constant throughout the life of the feeder)	: Rs.0.25 per kWh
12. Maximum allowable voltage drop, D	: 495 volts
13. Conductors available in the inventory	: Squirrel (20.95mm ² Al. equivalent), Weasel (31.63 mm ² Al. equivalent) & Rabbit (52.95 mm ² Al. equivalent).

The results obtained through the proposed method for the feeder in Fig.1 is as given below :

The optimum value of the objective function, $Z^* = \text{Rs.} 297\ 280$. The optimum conductor cross-section for the first seven line segments from the far end is squirrel, for the following middle three line segments is weasel and for the next three line segments up to the substation is rabbit. The same results were obtained through an enumeration technique also. The cost using a single optimal conductor cross-section, i.e. weasel in this case is Rs. 318 796 and the reduction in cost due to conductor gradation is Rs. 21 516.

The optimal results of the feeder in Fig.1 through the proposed algorithm have been obtained in about 6 seconds of cpu time on ICL 2960 computer system. The same problem when solved by an enumeration technique took about 25 seconds of cpu time on the same computer system.

Since the model proposed is independent of load distribution, load factor, load growth rate, cost of energy, investment cost, circuit voltage etc. on which the optimal conductor gradation is dependent, it is applicable to any type of radial feeder, may be overhead lines or underground cables in the rural or urban systems. Using the proposed method, the sensitivity of all the dependent variables on the conductor gradation can be analysed. The algorithm, suggested, considers load growth, growth in load factor and change in energy cost over the life of the feeder, minimizing the life time cost, thus giving the absolute optimal conductor gradation for the feeder. The proposed method is simple and easy to program and hence in the authors' opinion is quite promising.

NOMENCLATURE

I_i	phase current in Amps in the i th line-segment
R_i	per-phase resistance in ohms of the i th line-segment
X_i	per-phase reactance in ohms of the i th line-segment
x	per-phase reactance in ohms per km of a distribution line, assumed constant for a given range of conductor cross-sections.
l_i	length in km of the i th line-segment
v	line-line voltage drop of the feeder in volts
ρ	resistivity of the conductor material in ohm-mm ² /km
θ	power factor angle
LF_p	present load factor
LF_u	ultimate load factor
LF_k	load factor in the k th year
$(LLF)_k$	loss load factor in the k th year
g	annual load growth rate
r	annual discount rate
c_k	cost of energy in the k th year
c_f	feeder cost in Indian Rupees
D	maximum voltage drop allowable in the feeder
a_i	conductor cross-section of the i th line-segment

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APPENDIX A

Consider a simple radial distribution feeder as in Fig.a

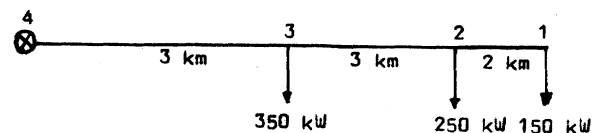


Fig. a A simple radial distribution feeder

with the following data :

Present load factor	= 0.20
Ultimate load factor	= 0.45
Annual rate of load growth	= 0.10
Load growth period	= 10 years
Life of the feeder	= 25 years
Annual discount rate	= 0.10
Cost of energy (assumed constant throughout feeder life)	= Rs.0.25 per kWh
Cost of line per mm ² /km	= Rs.360
Average reactance of the line	= 0.38 ohms/km
Maximum allowable voltage drop	= 414 volts
Conductors available in the inventory	= Squirrel, Weasel, Rabbit

With the above data, b in (19) works out to 314 volts. The objective and constraint functions for this problem can be stated as

$$\text{Min}_{a_1, a_2, a_3} Z = \left(\frac{166\ 412}{a_1} + 720\ a_1 \right) + \left(\frac{1\ 765\ 963}{a_2} + 1080\ a_2 \right) + \left(\frac{6\ 131\ 130}{a_3} + 1080\ a_3 \right) \quad (\text{A.1})$$

subject to

$$\frac{799}{a_1} + \frac{3190}{a_2} + \frac{5943}{a_3} \leq 314 \quad (\text{A.2})$$

and $a_1, a_2, a_3 > 0$ (each taking the values 20.95, 31.63 or 52.95).

Solution

The stage returns, $F_1(a_1)$ and the voltage drop contributions are as tabulated in Table A-1.

Table A-1

Stage returns and voltage drop contributions

a_1 (mm ²)	Stage 1		State 2		State 3	
	$F_1(a_1)$	d_1	$F_2(a_2)$	d_2	$F_3(a_3)$	d_3
20.95	23 027	38	106 920	152	315 281	284
31.63	28 035	25	89 992	101	227 999	188
52.95	41 267	15	90 538	60	172 977	112

The state variables,

$$S_3 = \frac{799}{a_1} + \frac{3190}{a_2} + \frac{5943}{a_3} \leq 314 \quad (\text{A.5})$$

$$S_2 = \frac{799}{a_1} + \frac{3190}{a_2} = S_3 - \frac{5943}{a_3} \quad (\text{A.6})$$

$$S_1 = \frac{799}{a_1} = S_2 - \frac{3190}{a_2} \quad (\text{A.7})$$

The minimum feasible value of S_1 , S_2 and S_3 are obtained from (26) as 15, 75, 187, respectively. The maximum value for these state variables is 314.

Stage 1 : The state function $F_1(S_1)$ and the corresponding policy function $\hat{a}_1(S_1)$ for all the feasible values of S_1 are developed as in Table A-2.

Table A-2

S_1	$F_1(a_1)$			$F_1(S_1)$	$\hat{a}_1(S_1)$
	$a_1=20.95$	$a_1=31.63$	$a_1=52.95$		
15 to 24	-	-	41 267	41 267	52.95
25 to 37	-	28 035	41 267	28 035	31.63
38 to 314	23 027	28 035	41 267	23 027	20.95

Stage 2 : Using the recursive relations in (28), the state function $F_2(S_2)$ and the policy function $\hat{a}_2(S_2)$ for all the feasible values of S_2 are obtained as tabulated in Table A-3.

Table A-3

S_2	$F_2(a_2) + F_1(S_1)$			$F_2(S_2)$	$\hat{a}_2(S_2)$
	$a_2=20.95$	$a_2=31.63$	$a_2=52.95$		
75 to 84	-	-	131 805	131 805	52.95
85 to 97	-	-	118 573	118 573	52.95
98 to 115	-	-	113 565	113 565	52.95
116 to 125	-	131 259	113 565	113 565	52.95
126 to 138	-	118 027	113 565	113 565	52.95
139 to 166	-	113 019	113 565	113 019	31.63
167 to 176 148 187	113 019	113 565	113 019	113 019	31.63
177 to 189 134 955	113 019	113 565	113 019	113 019	31.63
190 to 314 129 947	113 019	113 565	113 019	113 019	31.63

Stage 3 : At stage 3, the final stage, Z^* is given by $F_3(b)$ which is obtained from (28) as in Table A-4.

Table A-4

S_3	$F_3(a_3) + F_2(S_2)$			$F_3(S_3)$	$\hat{a}_3(S_3)$
	$a_3=20.95$	$a_3=31.63$	$a_3=52.95$		
314	-	341 564	285 996	285 996	52.95

The '-' in Tables A-2, A-3 and A-4 indicate the non-feasibility of such alternatives.

Thus

$$Z^* = F_3(314) = 285 996, \text{ and } a_3^* = 52.95 \text{ (Rabbit)}$$

a_1^* and a_2^* are obtained as follows :

$$S_2 = S_3 - \frac{5943}{a_3^*} = 314 - 112 = 202$$

$$\text{From } \hat{a}_2(S_2) \text{ in Table A-3, } a_2^* = \hat{a}_2(202) = 31.63 \text{ (Weasel)}$$

$$S_1 = S_2 - \frac{3190}{a_2^*} = 202 - 101 = 101$$

$$\text{From } \hat{a}_1(S_1) \text{ in Table A-2, } a_1^* = \hat{a}_1(101) = 20.95 \text{ (Squirrel)}$$

The optimal conductor grading for the feeder considered is obtained as Squirrel, Weasel and Rabbit for the first, second and third line segments, respectively, from the far end of the feeder.

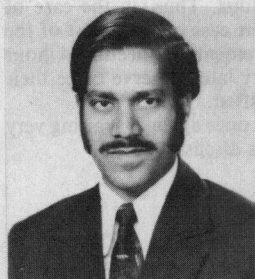
APPENDIX B

The main limitation in optimal conductor grading for radial distribution feeders is that the configuration of the feeder mains may be changed, once a new substation is installed in the system and in which case the optimal solution may be different. Also, when a new substation is installed, a remote portion of a feeder with least conductor cross-section may be transferred to the new substation which may form the forehead of a new feeder, necessitating reconductoring of some such sections. But such situations in a system are infrequent. Normally, substations, required for a period of at least 5 to 10 years are planned and installed in advance and the configuration of the feeder mains remains more or less constant during the plan period, say about 10 years. However, when there is a change in circuit configuration, the changed feeders need to be analysed for optimal conductor gradation and the results may be implemented through an optimal reconductoring schedule, to be evolved independently.

There may be some feeders in the system with capacitors installed on-line. The effect of such capacitors may be considered in the model by appropriately taking the changed power factor angles in the equations (5) and (6) for computing k_{1i} and k_{2i} and while computing the sectional current flows. If this aspect is to be considered in the model, equations (5) and (6) will get modified as in (B.1) and (B.2) respectively.

$$k_{1i} = \sqrt{3} I_i l_i \rho \cos \theta_i \quad (\text{B.1})$$

$$k_{2i} = \sqrt{3} I_i l_i \times \sin \theta_i \quad (\text{B.2})$$



M. Ponnaivaikko was born in Sengamedu, South Arcot, Tamil Nadu, India, on March 7, 1946. He received his B.E. (Elec.) and M.Sc.(Engg.) in Power Systems Engineering from the College of Engineering, Guindy Madras, India, in 1969 and 1972, respectively. He is presently working for his Ph.D. Degree at the Indian Institute of Technology, Delhi.

He joined the Planning Directorate of Tamil Nadu Electricity Board in 1972 and was on deputation at the Southern Regional Electricity Board, Bangalore from 1974 to 1976 on special duty for conducting power system planning studies. During 1976 and 1977, he was with the Bharat Heavy Electricals Ltd. He is now at the Rural Electrification Corporation Ltd. as Special Officer responsible for System Improvement Projects and Training of Engineering personnels on computer aided distribution system planning. He has been consultant to many electrical utilities in India. His fields of interests are Transmission and Distribution System Planning, Unit Commitment for load despatch and generator maintenance scheduling.

Discussion

R. B. Adler (Philadelphia Electric Co., Philadelphia, PA): The authors are to be complimented on a well-written and interesting paper. As the costs of energy and generating capacity increase, the mathematical formulation and solution of conductor sizing problems becomes increasingly relevant [8]. The authors' consideration of the following questions would be appreciated.

After considering various conductor gradation examples, can the authors make any generalization on the variables which have the most influence on optimal conductor sizes. Consider, for instance, the relative influences on conductor sizes of ultimate circuit load, rate at which this load develops, load factor, allowable voltage drop, and cost of energy. If any generalization is possible, could this lead to a simplification of the optimization problem, perhaps to the point of establishing general guidelines for conductor grading.

Have the authors found the voltage drop constraint to be binding on the optimal conductor sizes? With larger conductor sizes based on economic rather than thermal considerations, the limit on steady-state voltage drop may assume less importance. Transient motor-starting currents can cause irritating lamp flicker and may, in some cases, become a more binding constraint. Would the authors comment?

Primary circuits of the open-tie radial configuration are becoming the standard in the United States. This requires that primary laterals serve, on occasion, twice their normal peak load. With the economic conductor sizes that your optimization yields, are the 4-hour emergency thermal ratings sufficient, or are conductors excessively tapered?

Again the authors are to be congratulated on a well-written paper on a timely topic.

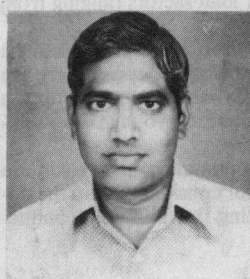
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- [8] W. G. Kirn and R. B. Adler, "A Distribution System Cost Model and Its Application to Optimal Conductor Sizing," presented at the IEEE Power Industry Computer Applications (PICA) Conference, May 1981.

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M. Ponnaivaikko and K. S. Prakasa Rao: The authors wish to thank the discussor, Dr. R. B. Adler, for his appreciation of the problem and for his very useful comments.

The question of generalization on optimal conductor sizing, as raised, becomes more relevant if one wants to provide a ready reckoner to the practising engineers for expanding their system more economically as the load grows. But some of the dependent variables are very sensitive and hence they do not permit generalization for establishing very general guidelines. For instance, load factor and load distribution have the maximum influence on optimal conductor grading. The load factor,



K.S. Prakasa Rao (S'70 -M'74) was born in Prakkilanka, Andhra Pradesh, India, in 1942. He received his B.E. (Elec.) and M.E. (Power Systems Engineering) Degrees from Osmania University, Hyderabad, India, in 1964 and 1966 respectively. In 1974, he obtained his Ph.D. Degree from the Indian Institute of Technology, Kanpur, India.

From 1966 to 1970, he was on the faculty of the Electrical Engineering Department of the Regional Engineering College, Warangal, Andhra Pradesh, India. From 1974 to 1975, he was with the University of Roorkee, Roorkee and from 1975 to 1978 he was a Professor in Power System Engineering at the National Institute of Engineering, Mysore, Karnataka, India. He joined the Centre of Energy Studies at the Indian Institute of Technology, New Delhi, India in 1978, where he is currently working on the Electrical Energy Systems Projects. His fields of interest are Load Flow Studies, Steady State Security Analysis and Optimal Operation of Inter-connected Power Systems.

in practice, varies from as low a value as 0.10 to as high a value as 0.60. Since the loss load factor, on which the loss cost is dependent, is a function of the square of the load factor, the effect due to a change in load factor, on optimal conductor grading, becomes more pronounced. Also, the load distribution on the actual feeders, is invariably nonuniform, each feeder differing from the other. The less cost of a feeder section is a function of the square of the load current in the section which is dependent on the feeder load distribution. Hence, any change in load distribution greatly influences optimal conductor grading. The changes in the other variables such as ultimate circuit load, rate of load growth, cost of energy and voltage drop have comparatively less influence on optimal conductor grading. The ultimate circuit load develops at a uniform rate at all the tapping points in proportion to the tapped loads, over a period of M years and remains constant for the rest of the period, during the life of the feeder, as per assumptions made in our model. The uniform growth in load along the feeder main keeps the relative increase in the loss costs in the different feeder sections also uniform and therefore, the grading problem is not much influenced by the changes in the rate of growth of load. However, the ultimate circuit load can be taken care of by suitably selecting the conductor sizes, from the inventory, for consideration in the optimization process. The effect of change in the cost of energy is of course considerable, if the changes occur over a wide range. Since the cost function in our model depends linearly on the cost of energy, the influence due to changes in cost of energy on optimal conductor grading is not as high as that of load factor or load distribution. The change in the allowable voltage drop has no control on optimal grading when its value exceeds that obtained purely from economic considerations. Variations in the allowable voltage drop within such economic levels, certainly influences the optimal grading. From our experience, it appears that we can only generalize in the case of uniformly loaded feeders. For a given load factor, the gradation of different conductor sizes along the feeder main can be expressed as a percentage of the total main length, or as a ratio of their lengths. For example, for a uniformly loaded feeder with a load factor of 0.2, the optimal grading with two or three conductor configuration can be generalized as given below:

S1. No.	Conductor Combination	Ratio of the lengths of the conductor sanction in the feeder main.
1.	Weasel and Squirrel (20mm ² (13mm ² ACSR) ACSR)	3 : 2
2.	Rabbit and Weasel (30mm ² ACSR)	2 : 3
3.	Rabbit and Squirrel	1 : 1
4.	Rabbit, Weasel and Squirrel	3 : 3 : 4

Similar generalized results are available for load factors ranging from 0.05 to 0.6.

As already mentioned above, the voltage drop constraint has got a definite binding on the optimal conductor grading in cases where the economic voltage drop is more than the allowable voltage drop. In other cases, as rightly pointed out by Dr. Adler, the voltage drop constraint assumes less importance.

The conductor gradation is subject to the minimum and maximum conductor sizes selected, from the inventory for consideration in the optimization process. Hence, by suitably selecting the minimum conduc-

tor size, the excessive tapering can be avoided. Thus, in the case of open radial configuration, which is also the case even in most of the developing countries like India, the requirement of meeting 4 hour emergency thermal ratings when the primary laterals serve twice their normal peak load, occasionally, can be satisfied.

The authors would like to thank Dr. Adler once again for raising very important and interesting points through his discussion.

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